MUTUAL GRAVITATIONAL POTENTIAL OF N SOLID BODIES

N. BORDERIES

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Abstract. The mutual gravitational potential of N solid bodies is expanded without approximation in terms of harmonic coefficients of each body. As an application the Euler dynamical equations for the motion of the axis of figure of the rigid Earth are integrated analytically by the method of variation of parameters.

1. Introduction

Let us consider N solid bodies in the solar system. The motion of any of them S consists of two component motions, namely the motion of its center of mass and a rotation around it, which are not dynamically independent of each other. These motions are perturbed by the gravitational attractions of the other bodies derived from a force function U which determines, together with the initial conditions, the dynamical equations of motion. Generally the expressions for U involve the interaction of the external bodies considered as point masses (or composed of homogeneous concentric spherical layers) with the distribution of mass of S. But this simplification may not be adequate for precise investigation.

Giacaglia and Jefferys (1971), and Bursa (1973) attempted to solve approximately this problem but their results, which come out as truncated series, cannot be generalized.

The purpose of this paper is to give a general and rigorous expansion for the mutual potential energy of N solid bodies. Moreover it will be seen that this new result allows us to construct a general theory for the motion of a solid body around its center of mass.

2. Gravitational Potential Energy of N Solid Bodies

Consider N solid bodies S_i in the solar system and let O_i be their center of mass, M_i their mass. Let $(\mathscr{R})$ be an absolute reference frame. (R_i) and (r_i) are two sets of coordinate axis of origin O_i: (R_i) is parallel to $(\mathscr{R})$; (r_i) is the principal axis system of the body S_i; the principal moments of inertia are A_i, B_i, C_i.

The potential energy of these N solid bodies is:

$$U = \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} U_{ij} \tag{1}$$

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with:

$$U_{ij} = G \int_{S_i} \int_{S_j} \frac{\mathrm{d}M_i \, \mathrm{d}M_j}{\rho_{ij}} \tag{2}$$

where G is the constant of gravitation, ρ_{ij} the distance between the elements of mass $\mathrm{d}M_i$ and $\mathrm{d}M_j$ around the current points P_i and P_j of S_i and S_j respectively.

We look for an explicit expression of U depending on the mass distribution of each of the bodies and on a set of variables specifying for each body the position of its center of mass as well as the rotation state around it.

Due to the form of (1) we can restrict ourselves to the case of N equal to 2:

$$U = \int_{S_2} \left[\int_{S_1} \frac{G\,dM_1}{\rho_{12}} \right] \mathrm{d}M_2. \tag{3}$$

We shall denote by R_i, ϕ_i, Λ_i the ordinary spherical coordinates relative to (R_i) and by R_0, ϕ_0, Λ_0 those of O_2 relative to (R_1). (Figure 1.)

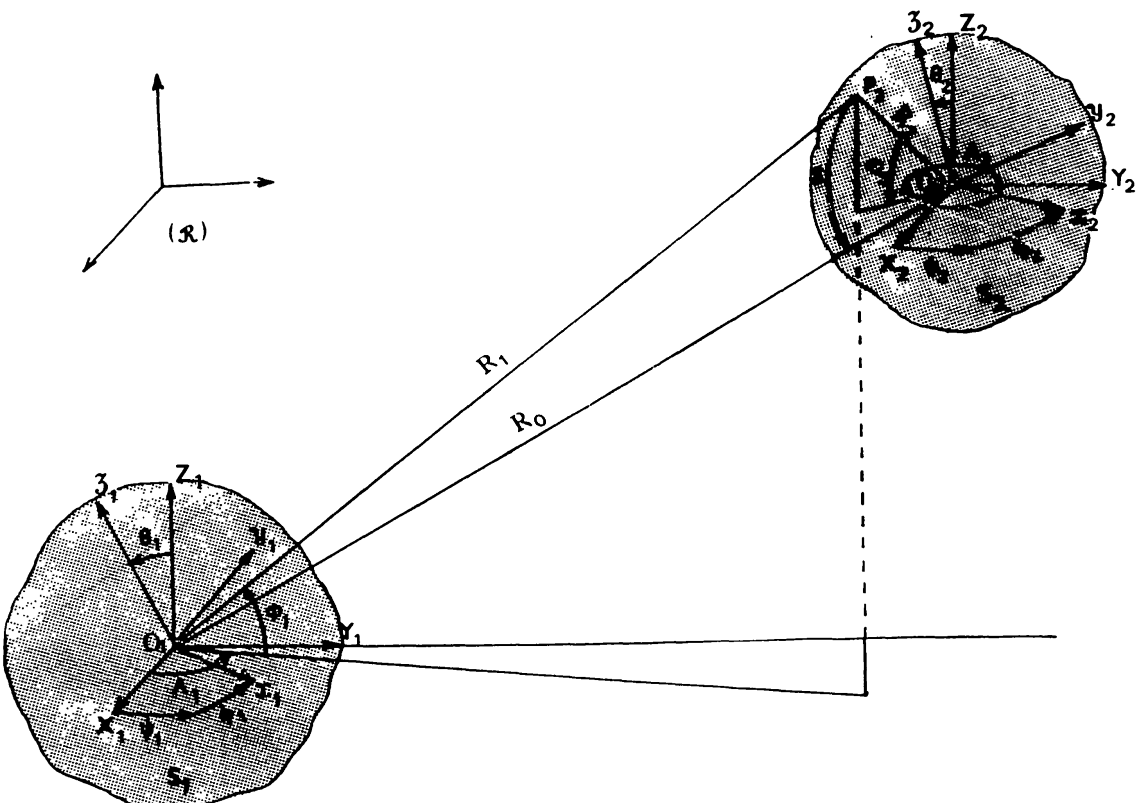

Fig. 1. Coordinate systems and Eulerian angles.

2.1. TRANSFORMATION OF THE EXPRESSION INTO BRACKETS

P_2 being fixed in S_2, the expression into brackets in (3) is the potential of S_1 at P_2. It can be written as follows:

$$\int_{S_1} \frac{G\,\mathrm{d}M_1}{\rho_{12}} = GM_1 \sum_{l=0}^{+\infty} \sum_{m=-l}^{l} \frac{a_{e1}^l}{R_1^{l+1}} K_{lm}^{1*} P_{lm}(\sin\phi_1)\, \mathrm{e}^{im\Lambda_1} \tag{4}$$

P_{lm} is the Legendre function:

$$\frac{P_{l,-k}}{(l-k)!} = (-1)^k \frac{P_{lk}}{(l+k)!}. \tag{5}$$

The K_{lm}^{1*} are moments of mass distribution of S_1. Subscript 1 refer to S_1 and star refer to (R_1). These coefficients are defined by:

$$K_{lm}^{1*} = \frac{(-1)^m}{M_1 a_{e1}^l} \int_{S_1} R_1'^l P_{l,-m}(\sin\phi_1')\, \mathrm{e}^{-im\Lambda_1'}\, \mathrm{d}M_1 \tag{6}$$

where a_{e1} is the radius of the smallest sphere containing the body S_1. The coefficients K_{lm}^{1*} are related to the usual C_{lm}^{1*} and S_{lm}^{1*} by:

$$K_{lm}^{1*} = \begin{cases} \dfrac{1+\delta_{0m}}{2}(C_{lm}^{1*} - iS_{lm}^{1*}) & \text{if } m \geq 0 \\[2ex] \dfrac{1+\delta_{0m}}{2}(C_{l,-m}^{1*} + iS_{l,-m}^{1*}) \dfrac{(l-m)!}{(l+m)!}(-1)^m & \text{if } m \leq 0. \end{cases}$$

We have obtained

$$\int_{S_1} \frac{G\,\mathrm{d}M_1}{\rho_{12}}$$

as a function F of R_1, ϕ_1, Λ_1, which we must integrate over the volume of S_2. Unfortunately such a function is not adapted to this integration. This difficulty will be overcome by expressing F in terms of the spherical coordinates R_2, ϕ_2, Λ_2 of P_2 relative to (R_2). So the next step is to transform $(1/R_1^{l+1})P_{lm}(\sin\phi_1)\,\mathrm{e}^{im\Lambda_1}$ under translation of coordinate axis.

2.2. TRANSFORMATION OF $(1/R_1^{l+1})P_{lm}(\sin\phi_1)\,\mathrm{e}^{im\Lambda_1}$ UNDER TRANSLATION OF COORDINATE AXIS

We start from:

$$\frac{1}{R_1} = \sum_{j=0}^{+\infty} \frac{R_2{}^j}{R_0{}^{j+1}} P_j(\cos\mathscr{S}) \tag{8}$$

where $\mathscr{S}$ is the angle between O_2P_2 and O_2O_1.

Using the addition theorem of spherical functions the above expression can be put under the following form:

$$\frac{1}{R_1} = \sum_{j=0}^{+\infty} \sum_{k=-j}^{j} (-1)^j \frac{(j-k)!}{(j+k)!} P_{jk}(\sin\phi_0)\, \mathrm{e}^{ik\Lambda_0} \frac{R_2{}^j}{R_0{}^{j+1}} P_{jk}(\sin\phi_2)\, \mathrm{e}^{-ik\Lambda_2}. \tag{9}$$

We have obtained $1/R_1$ in terms of R_2, ϕ_2, Λ_2. But how to transform the whole expression $(1/R_1^{l+1})P_{lm}(\sin\phi_1)\,e^{im\Lambda_1}$ is not at all obvious.

The following step is fundamental to work out the problem of determining the potential energy of the system S_1, S_2. The basic relation is:

$$\frac{1}{R_1^{l+1}}P_{lm}(\sin\phi_1)\,e^{im\Lambda_1} = \frac{(-1)^l}{(l-m)!}\,A_1^{l-m}A_2^{l+m}\left(\frac{1}{R_1}\right) \tag{10}$$

where the operators A_1 and A_2 are defined by:

$$\left.\begin{aligned} A_1^2 &= -\frac{\partial}{\partial X_1} + i\frac{\partial}{\partial Y_1} \\ A_2^2 &= \frac{\partial}{\partial X_1} + i\frac{\partial}{\partial Y_1} \end{aligned}\right\} \tag{11}$$

Note that since $X_1 = X_0 + X_2, \ldots,$ we have $\partial/\partial X_1 = \partial/\partial X_0, \ldots.$

Now the method is as follows: First, the basic relation (10) is applied to $(1/R_0^{j+1})$ $P_{jk}(\sin\phi_0)\,e^{ik\Lambda_0}$ in the right member of (9). Afterwards the transformed expression for $1/R_1$ is substituted into the right member of relation (10). Then the obtained expression is transformed by using again the basic relation (10). One has finally:

$$\frac{1}{R_1{}^{l+1}}\,P_{lm}(\sin\phi_1)\,e^{im\Lambda_1} = \sum_{j=0}^{+\infty}\sum_{k=-j}^{j}\gamma_{lm}^{jk}R_2{}^{j}P_{j,-k}(\sin\phi_2)\cdot$$

$$e^{-ik\Lambda_2}\frac{1}{R_0{}^{l+j+1}}P_{l+j,\,m+k}(\sin\phi_0)\,e^{i(m+k)\Lambda_0} \tag{12}$$

with:

$$\gamma_{lm}^{jk} = (-1)^{j+k}\,\frac{(l+j-m-k)!}{(l-m)!\,(j-k)!}$$

2.3. INTEGRATION OVER S_2

The previous result allows to obtain

$$\int_{S_1}\frac{G\,dM_1}{\rho_{12}}$$

as a function $\tilde{F}$ of R_2, ϕ_2, Λ_2 and it remains to integrate $\tilde{F}$, or rather $R_2^jP_{j,\,-k}(\sin\phi_2)$ $e^{-ik\Lambda_2}$, over the volume of S_2. It turns out that one obtains a very pleasant result namely that the integration gives the moments of mass distribution K_{jk}^{2*} of S_2 relative to (R_2). Finally:

$$U = GM_1M_2\sum_{l=0}^{+\infty}\sum_{m=-l}^{l}\sum_{j=0}^{+\infty}\sum_{k=-j}^{j}\delta_{lm}^{jk}\frac{a_{e1}^{l}a_{e2}^{j}}{R_0{}^{l+j+1}}P_{l+j,m+k}\,(\sin\phi_0)\times$$

$$\times\, e^{i(m+k)\Lambda_0}\,K_{lm}^{1*}K_{jk}^{2*} \tag{13}$$

where:

$$\delta_{lm}^{jk} = (-1)^k \gamma_{lm}^{jk} = (-1)^j \frac{(l+j-m-k)!}{(l-m)!\,(j-k)!}.$$

Let us recall that in (13) the moments of mass distribution K_{np}^{i*} refer to a system of coordinates moving in S_i and obviously they vary with time. It appears that the choice of harmonic coefficients K_{np}^{i} relative to the principal axis system (r_i) is more convenient since they are constant with time provided that the body is rigid. This choice allows us further to apply time variations to these coefficients for a non-rigid model.

2.4. TRANSFORMATION OF HARMONIC COEFFICIENTS UNDER ROTATION OF COORDINATE AXIS

The relation between harmonic coefficients in (R_i) and (r_i) depends on the Euler angles ψ_i, θ_i, φ_i orientating (r_i) with respect to (R_i), and results from the theorem of Wigner (1959) about the transformation of spherical functions under rotation of coordinate axis: α, β (respectively α', β') being the longitude and latitude relative to (R_i) (respectively (r_i)) one has:

$$P_{lm}(\sin\beta)\,\mathrm{e}^{im\alpha} = \frac{1}{(l-m)!} \sum_{m'=-l}^{l} (l-m')!\,E_{lm}^{m'}(\psi_i, \theta_i, \varphi_i) \times$$
$$\times\, P_{lm'}(\sin\beta')\,\mathrm{e}_i^{m'\alpha'} \tag{14}$$

where the Euler functions $E_{lm}^{m'}$ are defined by:

$$E_{lm}^{m'}(\psi\,\theta, \varphi) = (-1)^{l-m}\,\mathrm{e}^{i(m'-m)(\pi/2)}\,\mathrm{e}^{i(m\psi+m'\varphi)} H_{lm}^{m'}(\theta) \tag{15}$$

with:

$$H_{lm}^{m'}(\theta) = \sum_{j=\sup(0,-m-m')}^{\inf(l-m,l-m')} (-1)^j \binom{l-m}{j} \binom{l+m}{m+m'+j} \times$$
$$\times \left(\cos\frac{\theta}{2}\right)^{2j+m+m'} \left(\sin\frac{\theta}{2}\right)^{2(l-j)-m-m'}. \tag{16}$$

It follows that:

$$K_{lm}^{i*} = \frac{1}{(l+m)!} \sum_{m'=-l}^{l} (-1)^{m+m'} (l+m')!\; K_{lm'}^{i} E_{l,-m}^{-m'}(\psi_i, \theta_i, \varphi_i). \tag{17}$$

2.5. FINAL EXPRESSION OF THE GRAVITATIONAL POTENTIAL

Substituting (17) in (13) one obtains:

$$U = GM_1M_2 \sum_{l=0}^{+\infty} \sum_{j=0}^{+\infty} \sum_{m,m'=-l}^{l} \sum_{k,k'=-j}^{j} (-1)^{j+m+m'+k+k'} \times$$

$$\times\ \alpha_{ljmm'kk'} \frac{a_{e1}^{l} a_{e2}^{j}}{R_0^{l+j+1}} P_{l+j,m+k}(\sin\phi_0) \times$$

$$\times\ e^{i(m+k)\Lambda_0} K_{lm'}^{1} K_{jk'}^{2} E_{l,\,-m}^{-\ m'}(\psi_1, \theta_1, \varphi_1) \times \tag{18}$$

$$\times\ E_{j,\,-k}^{-\ k'}(\psi_2, \theta_2, \varphi_2)$$

with:

$$\alpha_{ljmm'kk'} = \frac{(l+j-m-k)!\,(l+m')!\,(j+k')!}{(l-m)!\,(l+m)!\,(j-k)!\,(j+k)!}.$$

As expected, the gravitational potential U is expanded in terms of harmonic coefficients of each body and involves 9 variables, namely R_0, ϕ_0, Λ_0 which determine the relative position of the centers of mass and for $i = 1, 2$, ψ_i, θ_i, φ_i which specify the rotation state of S_i around O_i.

2.6. SOME REMARKS

Using formula (18) calls for some remarks. The force function U (or its generalization as indicated by (1)) can be introduced in the right members of the dynamical equations describing the motion of N bodies. In fact the problems in which one is generally involved are more particular than that of studying the whole motion of the whole system. Anyhow, in order to work out a given problem one is led to choose a formalism such that a convenient change of variables may be required. In practice two kinds of configurations arise: S_2 is in orbit around S_1; or S_1 and S_2 are in orbit around a third body. Often adequate new variables are elliptic elements instead of R_0, ϕ_0, Λ_0. In the first type of problem the transformation of U can be achieved using classical Kaula (1966) functions F_{lmp} and G_{lpq} of the orbital inclination and the eccentricity respectively:

Let a_2, e_2, I_2, Ω_2, ω_2, M_2 be the elliptic elements of O_2 relative to (R_1). One obtains:

$$U = GM_1M_2 \sum_{l=0}^{+\infty} \sum_{j=0}^{+\infty} \sum_{m,m'=-l}^{l} \sum_{k,k'=-j}^{j} \beta_{ljmm'kk'} \times$$

$$\times\ \frac{a_{e1}^{l} a_{e2}^{j}}{a^{l+j+1}} K_{lm'}^{1} K_{jk'}^{2} H_{lm}^{m'}(\theta_1) H_{jk}^{k'}(\theta_2) \sum_{p=0}^{l+j} \sum_{q=-\infty}^{+\infty} \times$$

$$\times\ D_{l+j,m+k,p}(I_2) G_{l+j,p,q}(e_2)\, e^{iA_{ljmm'kk'}^{pq}} \tag{19}$$

with the argument:

$$A_{ljmm'kk'}^{pq} = (l+j-2p)\omega_2 + (l+j-2p+q)M_2 + (m+k)\Omega_2 -$$

$$- m\psi_1 - k\psi_2 - k'\varphi_2 - m'\varphi_1$$

and with:

$$\beta_{ljmm'kk'} = (-1)^{l+m+k} i^{l+j-m'-k'} \alpha_{ljmm'kk'}$$

$$D_{lmp}(I_2) = \begin{cases} (-1)^{[(l-m)/2]+(l-m)} F_{lmp}(I_2) & \text{if } m \geq 0 \\ (-1)^{[(l-m)/2]+m} \dfrac{(l+m)!}{(l-m)!} F_{l,-m,l-p}(I_2) & \text{if } m \leq 0. \end{cases} \tag{20}$$

On the other hand in the second type of problem, the change of spherical elements into elliptic elements can be achieved only after having transformed $(1/R_o^{l+j+1})$ $P_{l+j,m+k}(\sin \phi_0)\, e^{i(m+k)\Lambda_0}$ by means of the translation formula (12):

Let $(a_i, e_i, I_i, \Omega_i, \omega_i, M_i)$ be the elliptic elements of O_i $(i = 1, 2)$ relative to a reference frame parallel to $(\mathscr{R})$ and the origin of which coincides with the center of mass of the third body. We assume further that $a_1 > a_2$. One obtains:

$$U = GM_1M_2 \sum_{l=0}^{+\infty} \sum_{j=0}^{+\infty} \sum_{m,m'=-l}^{l} \sum_{k,k'=-j}^{j} \sum_{r=0}^{+\infty} \sum_{s=-r}^{r} \gamma_{ljmm'kk'}^{rs} \times$$

$$\times \frac{a_{e1}^l a_{e2}^j}{a_1^{l+j+1}} \left(\frac{a_2}{a_1}\right)^r K_{lm'}^1 K_{jl'}^2 H_{lm}^{m'}(\theta_1) H_{jk}(\theta_2) \times$$

$$\times \sum_{p=0}^{l+j+r} \sum_{q=-\infty}^{+\infty} D_{l+j+r,m+k-s,p}(I_1) G_{l+j+r,p,q}(e_1) \times$$

$$\times \sum_{c=0}^{r} \sum_{d=-\infty}^{+\infty} D_{rsc}(I_2) H_{rcd}(e_2)\, e^{iB_{ljmm'kk'}^{rspqcd}} \tag{21}$$

with the argument:

$$\begin{aligned} B_{ljmm'kk'}^{rspqcd} = {} & (r-2c)\omega_2 + (r-2c+d)M_2 + s\Omega_2 + (l+j+r-2p)\omega_1 + \\ & + (l+j+r-2p+q)M_1 + (m+k-s)\Omega_1 - m\psi_1 - \\ & - m'\varphi_1 - k\psi_2 - k'\varphi_2 \end{aligned}$$

and:

$$\begin{aligned} \lambda_{ljmm'kk'}^{rs} = {} & (-1)^{j+m+k+r+s}\, i^{l+j-m'-k'} \times \\ & \times \frac{(l+j-m-k+r+s)!\,(l+m')!\,(j+k')!}{(l-m)!\,(l+m)!\,(j-k)!\,(j+k)!\,(r+s)!} \end{aligned}$$

H_{rcd} is Hansen function.

3. Application to the Rotation of Solid Bodies in the Solar System

The formalism which has been developed is very useful for studying the rotation of solid bodies perturbed by gravitational attractions of other bodies. For instance it allows investigations about precession and nutation of the Earth.

3.1. METHOD

Let S_1 be the Earth and S_2 the Moon. In order to simplify the presentation we make the following hypothesis:

(1) The Moon is a point mass (or composed of homogeneous, concentric, spherical layers); then $j = 0$.

(2) We restrict ourselves to the term of degree of 2 of the force function: $l = 2$.

(3) Moreover we take $A = B$ such that all the harmonic coefficients of degree 2 are equal to zero except $C_{20} : C_{20} = -(1/M_1 a_{e1}^2)/(C - A)$; then $m' = 0$.

The force function is in terms of elliptic elements of the Moon:

$$U_2 = -\frac{GM_2(C-A)}{a^3}2\sum_{m=0}^{2}\sum_{p=0}^{2}\sum_{q=-\infty}^{+\infty}(-1)^{m+[m/2]}\times$$
$$\times\frac{2-\delta_{0m}}{(2+m)!}H_{2m}^0(\theta)F_{2mp}(I)G_{2pq}(e)\tag{22}$$
$$\times\cos[(2-2p)\omega+(2-2p+q)M+m(\Omega-\psi)]$$

Let p, q, r be the components of the angular velocity vector along the principal axis:

$$\left.\begin{aligned} p &= \dot\psi\sin\theta\sin\phi+\dot\theta\cos\phi\\ q &= \dot\psi\sin\theta\cos\phi-\dot\theta\sin\phi\\ r &= \dot\psi\cos\theta+\dot\phi.\end{aligned}\right\}\tag{23}$$

The motion of the rigid Earth relative to its center of mass is represented by the Euler dynamical equations which, with $A = B$, reduce to:

$$\left.\begin{aligned} \dot p+\frac{C-A}{A}qr &= \frac{\sin\phi}{A\sin\theta}\frac{\partial U_2}{\partial\psi}+\frac{\cos\phi}{A}\frac{\partial U_2}{\partial\theta}\\ \dot q-\frac{C-A}{A}pr &= \frac{\cos\phi}{A\sin\theta}\frac{\partial U_2}{\partial\psi}-\frac{\sin\phi}{A}\frac{\partial U_2}{\partial\theta}\\ \dot r\qquad\qquad &= 0.\end{aligned}\right\}\tag{24}$$

We integrate these equations by the method of variation of parameters. This leads to a solution of the form:

$$\left.\begin{aligned} p &= f\cos\mu t+g\sin\mu t\\ q &= f\sin\mu t-g\cos\mu t\\ r &= \text{const.}\end{aligned}\right\}\tag{25}$$

with

$$\mu = \frac{C-A}{A}r.\tag{26}$$

One has the following differential equations for f and g:

$$\left.\begin{aligned}\dot{f} &= \frac{\sin(\phi + \mu t)}{A \sin\theta}\frac{\partial U_2}{\partial\psi} + \frac{\cos(\phi + \mu t)}{A}\frac{\partial U_2}{\partial\theta}\\ \dot{g} &= -\frac{\cos(\phi + \mu t)}{A\sin\theta}\frac{\partial U_2}{\partial\psi} + \frac{\sin(\phi + \mu t)}{A}\frac{\partial U_2}{\partial\theta}.\end{aligned}\right\} \tag{27}$$

It follows from (23) that:

$$\left.\begin{aligned}\dot{\psi} &= [f\sin(\phi + \mu t) - g\cos(\phi + \mu t)]/\sin\theta\\ \dot{\theta} &= f\cos(\phi + \mu t) + g\sin(\phi + \mu t)\\ \dot{\phi} &= r - \dot{\psi}\cos\theta.\end{aligned}\right\} \tag{28}$$

Then systems (27) and (28) are solved analytically by successive approximations. To zero order one obtains with the secular part of U_2 ($m = 0$, $p = 1$, $q = 0$) the lunar precession rate:

$$\dot{\psi} = -\frac{3}{2}\frac{GM_2}{a^3 r}\frac{C - A}{C}\cos\theta(1 - \tfrac{3}{2}\sin^2 I)(1 - e^2)^{-3/2}. \tag{29}$$

For the approximation to first order we take ψ and ϕ linear, θ constant, and a secularly precessing Kepler ellipse for the Moon. We derive the variations corresponding to the term U_{2mpq} of the force function:

$$U_{2mpq} = VH\cos u$$

with

$$V = -\frac{GM_2(C - A)}{a^3}\, 2\frac{(2 - \delta_{0m})}{(2 + m)!}(-1)^{m + [m/2]}F_{2mp}(I)G_{2pq}(e)$$

$$H = H^0_{2m}(\theta)$$

$$u = (2 - 2p)\omega + (2 - 2p + q)M + m(\Omega - \psi). \tag{30}$$

The obtained variations are the nutations in longitude and obliquity:

$$\Delta\psi = \frac{V}{A\sin\theta}\frac{1}{\beta^2 - \dot{u}^2}\left(H'\frac{\beta}{\dot{u}} + \frac{mH}{\sin\theta}\right)\sin u \tag{31}$$

$$\Delta\theta = \frac{V}{A}\frac{1}{\beta^2 - \dot{u}^2}\left(H' + \frac{mH}{\sin\theta}\frac{\beta}{\dot{u}}\right)\cos u. \tag{32}$$

H' is the derivative of H with respect to θ.

$$\beta = \dot{\phi} + \mu = \frac{C}{A}r - \dot{\psi}\cos\theta.$$

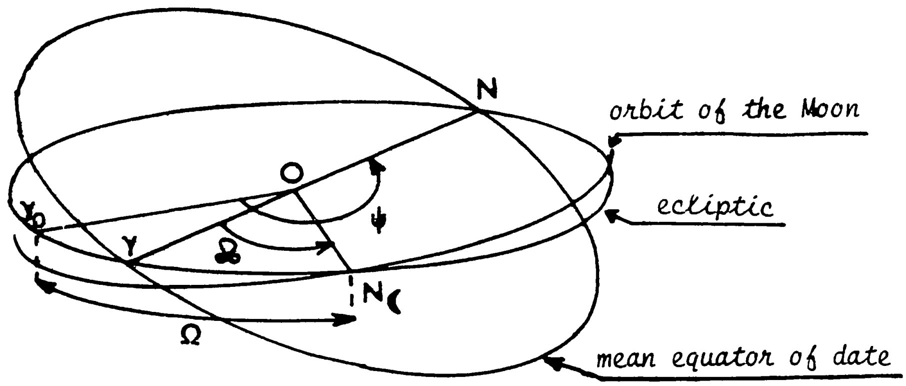

Fig. 2. Woolard notations (the tropic longitude of the ascending node is $\text{☊} = \Omega - \psi + \pi$; the tropic mean longitude of the Moon is $\text{☾} = \text{☊} + \omega + M$).

According to the notations adopted by Woolard (1953) and as shown in Figure 2, let ☾ be the tropic mean longitude of the Moon and ☊ the tropic longitude of the ascending node of the orbit; these longitudes refer to the ecliptic of epoch and to the corresponding mean equinox. Let us adopt, as Woolard, the retrograde sense as positive for the variations of ψ. Putting $\nu = (2 - 2p)\,\text{☾} + qM + (m - 2 + 2p)\text{☊}$, relations (31) and (32) become:

$$\Delta\psi = -(-1)^m \frac{V}{A \sin\theta} \frac{1}{\beta^2 - \dot{u}^2} \left(H' \frac{\beta}{\dot{u}} + \frac{mH}{\sin\theta} \right) \sin\nu \tag{33}$$

$$\Delta\theta = (-1)^m \frac{V}{A} \frac{1}{\beta^2 - \dot{u}^2} \left(H' + \frac{mH}{\sin\theta} \frac{\beta}{\dot{u}} \right) \cos\nu. \tag{34}$$

3.2. NUMERICAL RESULTS

Table I shows the values of lunar nutation coefficients for the principal and semi-monthly terms corresponding respectively to arguments ☊ and 2☾; these coefficients refer to the fixed ecliptic of 1900 Jan. 0, Greenwich mean noon. In evaluating the nutation coefficients we have used a set of constants which are given in the *Connaissance des Temps* published by the 'Bureau des Longitudes'.

A number of them are directly IAU (1964) astronomical constants:

Defining constant:	
Number of seconds of the ephemerides time in a tropical year (1900)	31 556 925.9747
Primary constants:	
Geocentric gravitational constant	$398\,603 \times 10^9$ m^3 s^{-2}
Ratio of mass of Moon to that of Earth	1/81.30
General precession in longitude per tropical century at epoch 1900.0	5025.″64

Obliquity of the ecliptic at epoch 1900,0	23° 27′ 8″.26
Sideral mean motion of the Moon at epoch 1900.0	$2.661\,699\,489 \times 10^{-6}$ rd s^{-1}

Secondary constant:

Perturbed semi-major axis of the orbit of the Moon	384 400 000 m

The other elements of the orbit of the Moon are from E. W. Brown, with some corrections so as to make them consistent with the IAU system of astronomical constants.

Sideral diurnal mean motion of Perigee	6′ 40″.917 094
Sideral diurnal mean motion of the node	−3′ 10″.771 714
Eccentricity	0.054 900 489
Inclination on the ecliptic	5° 8′ 43″.427

According to Le Verrier the Earth rotation period is taken as: 86 164.0990 s.

The value of $(C - A)/C$ compatible with the IAU constants is: 3.2732×10^{-3}.

Table I shows also the values of Woolard (1953) coefficients and weighted means of astronomical determinations given by McCarthy, Seidelmann and Van Flandern (1977).

TABLE I

Nutations coefficients (in seconds of arc)

	Term	Woolard	Our values	Astromical determinations (means)
Principal	Obliquity	9.2100	9.2237	9.2050
	Longitude	−17.2327	−17.2584	−17.1888
Semi-monthly	Obliquity	0.0884	0.0950	0.0925
	Longitude	− 0.2037	− 0.2215	− 0.2143

For comparing these values it is necessary to make some remarks:

(1) The Euler dynamical equations give the motion of the axis of figure of the Earth (about which the moment of inertia is maximum), while the values of Woolard and the astronomical determinations refer to the instantaneous rotational axis.

(2) The constants which determine the numerical values for lunar nutation appear in a single common factor of all the terms. The value of this constant factor is usually determined from the principal coefficient of nutation in obliquity, called the constant of nutation. Woolard has adopted the value 9.2100 based on observational results. In the contrary our value is really computed.

(3) There are discrepancies between astronomical observations and theoretical values for a rigid Earth. Also the ratio of these two kinds of values is not the same for several nutations such that a change of the constant of nutation is not sufficient. This has led several authors (Jeffreys and Vicente, 1957a and 1957b) to consider a non rigid Earth and it has been shown that core-mantle interactions and elasticity of the Earth affect the amplitudes of Lunar nutation.

3.3. IMPROVEMENTS

Nevertheless the rigid Earth theory of nutation is open to improvements and our formalism should permit some of them:

(1) Nutations due to other terms of the perturbing function can be derived using the same method as for U_2. In particular it is possible to calculate the variations corresponding to any harmonic coefficient of the Earth as well as the effect of Lunar ellipticity.

(2) The case $A \neq B$ can be treated by a method of perturbation as follows. Defining the small parameter ε by: $\varepsilon = (B - A)/C$ and putting:

$$\eta = \frac{1}{2}\left(\frac{C - A}{B} + \frac{C - B}{A}\right)$$

$$\alpha = \frac{C}{2AB}$$

$$\gamma = \alpha(C - A - B)$$

$$\lambda = \frac{A + B}{2AB}$$

The Euler dynamical equations can be written under the form:

$$\left.\begin{aligned} \dot{p} &= \lambda L - \eta qr + \varepsilon(\alpha L - \gamma qr) \\ \dot{q} &= \lambda M - \eta pr - \varepsilon(\alpha M + \gamma pr) \\ \dot{r} &= \varepsilon(N' - pq) \end{aligned}\right\} \tag{35}$$

where L, M, N are the components along the principal axis of the gravitational couple exerted on the Earth. One shows that N/C is of order ε: $N/C = \varepsilon N'$. Then the method consists in looking for a solution of the form:

$$\left.\begin{aligned} p &= (f_0 + \varepsilon f_1) \cos \mu t + (g_0 + \varepsilon g_1) \sin \mu t \\ q &= (f_0 + \varepsilon f_1) \sin \mu t - (g_0 + \varepsilon g_1) \cos \mu t. \end{aligned}\right\} \tag{36}$$

(3) On the other hand, further successive approximations of (27) and (28) can be achieved by using a program processing algebraic expressions on a computer, allowing in particular to best represent the Lunar orbital motion.

4. Conclusion

The fundamental result of this paper is the rigorous expansion of the gravitational potential energy of N solid bodies. We have seen that this formalism can be applied to study the nutation of the Earth. But more generally it allows, with similar methods, investigations of rotational variations of solid bodies in the solar system such as libration of the Moon and large oscillations of the obliquity of Mars under gravitational perturbations due to the other planets.

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